

Geofencing – how the server decides if the bus is on the right route

THE IDEA

We draw an invisible 'corridor' around the planned route (say 50 m wide on either side of the road).

If the bus stays inside the corridor → on route.
If it goes outside for several updates in a row → diverted.

THE MATH (used to check this)

Distance between two GPS points
(Haversine formula)

Tells us 'how far is the bus from this point on the route?'
Accurate to a few metres – well under GPS error.

Perpendicular distance from bus to a road segment
(Cross-track distance)

Tells us 'how far off the road line is the bus?'
This is the number we compare to the corridor width.

Fast version (good enough at our scale)

Convert lat/lon to flat x/y near the route, then standard high-school geometry to find the shortest distance from bus to road line.

~10× cheaper to compute, sub-metre error.

THE RULES (what the server actually does)

ON-ROUTE

Bus is within the corridor
(≤ 50 m from the road line)

ARRIVED at a stop

Bus is within 30 m of the stop AND barely moving

LEFT the stop

Bus is more than 50 m from the stop AND moving (extra 20 m so it doesn't flicker)

DIVERTED (off route)

Bus is outside the corridor for 3 packets in a row
(one bad GPS reading isn't enough – we wait for 3)

When diverted:

- log an alert in the database
- ping Person 3's dispatcher to notify passengers
- flash a banner on the authority dashboard